

Comparative Forecasting Performance of SARIMA, ETS, and NNAR Models for the Consumer Price Index of Fruits and Vegetables in Albania

Ilir Palla

Faculty of Natural and Human Sciences, Fan S. Noli University, Korçë, Albania

Jasmina Duka

Faculty of Natural and Human Sciences, Fan S. Noli University, Korçë, Albania

Abstract

This study aimed to identify the optimal model for forecasting the Consumer Price Index (CPI) for the fruits and vegetables category in Albania, while analyzing its seasonal dynamics and short-term volatility. The methodology relied on comparing three time series forecasting models: SARIMA, ETS, and NNAR. Forecast accuracy was evaluated using statistical measures such as MAPE and ACF1, while uncertainty was assessed through 95% confidence intervals. The results demonstrated that the SARIMA model exhibited the best performance for both series, achieving the lowest error values (MAPE = 2.38% for fruits and 3.76% for vegetables) and effectively eliminating autocorrelation in the residuals (ACF1 = 0.021 for fruits and ACF1 = - 0.018 for vegetables). The comparative analysis revealed that vegetables displayed higher seasonal volatility, with the point forecast peaking in March (178.38), compared to fruits, which peaked in June (143.04). The study confirmed that the vegetable market carries a higher forecasting risk and requires wider confidence intervals due to its sensitivity to supply-side factors. These findings suggest that the SARIMA model serves as a reliable tool for monitoring food inflation and guiding economic policy decisions.

Keywords: Consumer price index, SARIMA, Time series forecasting, Fruits and Vegetables

Introduction

The Consumer Price Index (CPI) is one of the most important macroeconomic indicators, measuring the average change in prices paid by households for a fixed basket of goods and services over time (International Labour Organization, 2004). In Albania, the fruits and vegetables category constitutes a significant share of the food basket and exhibits strong seasonality and short-term volatility due to weather conditions, harvest cycles, and supply shocks. Accurate forecasting of this sub-index is essential for monitoring food inflation, supporting monetary policy decisions by the Bank of Albania, and assisting producers and policymakers in planning. Understanding the methodological foundation of CPI construction is essential for interpreting its forecasting behaviour and selecting appropriate time series models.

CPI is typically constructed using the Laspeyres index formula, which compares the cost of a fixed basket of goods in the current period with the cost in the base period (International Labour Organization, 2004):

$$CPI_L = \frac{\sum_{i=1}^n P_{i,t} \times Q_{i,0}}{\sum_{i=1}^n P_{i,0} \times Q_{i,0}} \times 100 \quad (1)$$

where $P_{i,t}$ and $P_{i,0}$ are the prices of item in the current and base periods, respectively, and $Q_{i,0}$ is the quantity consumed in the base period. Although widely used for its simplicity, the Laspeyres index has limitations, such as not fully reflecting changes in consumption patterns or product quality.

Because prices observed at different times are not independent but display autocorrelation, CPI series are treated as time series data (Box et al., 2015; Enders, 2014). A time series is a sequence of observations of a variable, recorded in chronological order and typically at equal intervals, such as daily, monthly, or annually (Hyndman & Athanasopoulos, 2021). Such a series can be decomposed into four main components: trend (long-term movement), seasonality (regular periodic fluctuations, e.g., monthly), cycle (longer-term fluctuations), and irregular (random) residuals (Makridakis et al., 1998). For reliable forecasting, the series must satisfy the stationarity condition - that is, its statistical properties (mean, variance, and autocorrelation) remain constant over time (Enders, 2014). Stationarity is usually tested using the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test (Kwiatkowski et al., 1992). Autocorrelation and partial autocorrelation functions (ACF and PACF) further help identify the underlying dependence structure in the data (Box et al., 2015).

To forecast the Albanian CPI for fruits and vegetables, this study employs and compares three widely used time series models. The first is SARIMA(p,d,q)(P,D,Q)_m - Seasonal Autoregressive Integrated Moving Average - which extends the classical ARIMA framework by explicitly incorporating seasonal components, making it particularly effective for economic series exhibiting strong monthly or quarterly seasonality (Box et al., 2015; Hyndman & Athanasopoulos, 2021). The general SARIMA(p,d,q)(P,D,Q)_m model is expressed as (Box et al., 2015; Hyndman & Athanasopoulos, 2021):

$$\Phi_p(B^m)\phi_p(B)(1-B)^d(1-B^m)^D y_t = \Theta_Q(B^m)\theta_q(B)\varepsilon_t \quad (2)$$

where $\phi_p(B)$ and $\Phi_p(B^m)$ are the non-seasonal and seasonal autoregressive operators of orders p and P; $\theta_q(B)$ and $\Theta_Q(B^m)$ are the non-seasonal and seasonal moving average operators of orders q and Q; $(1-B)^d$ and $(1-B^m)^D$ denote non-seasonal and seasonal differencing of orders d and D; m is the seasonal period (m=12 for monthly data); and $\varepsilon_t \sim N(0, \sigma^2)$ is white noise.

The second model, ETS (Error-Trend-Seasonality), belongs to the exponential smoothing family and decomposes the series into level, trend, and seasonal components, either in additive or multiplicative form. The multiplicative specification performs particularly well when the amplitude of seasonality changes proportionally with the level of the series (Hyndman et al., 2008). For the additive specification ETS(A,A,A), the state space equations are defined as follows (Hyndman et al., 2008):

$$\text{Level: } l_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1}) \quad (3)$$

$$\text{Trend: } b_t = \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1} \quad (4)$$

$$\text{Seasonal: } s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m} \quad (5)$$

$$\text{Forecast: } y_t + h = l_t + h \cdot b_t + s_{t+h-m} \quad (6)$$

where y_t is the observed value at time t ; l_t is the level; b_t is the trend; s_t is the seasonal component; $\alpha \in [0,1]$, $\beta \in [0,1]$ and $\gamma \in [0,1]$ are the smoothing parameters for level, trend, and seasonality respectively; m is the seasonal period; and h is the forecast horizon. The optimal ETS specification was selected automatically by minimizing the corrected Akaike Information Criterion (AICc) (Hyndman et al., 2008).

The third approach, NNAR (Neural Network AutoRegressive), is a machine-learning method based on artificial neural networks, capable of capturing complex nonlinear relationships that linear models may fail to detect, making it a valuable benchmark in this comparative analysis (Zhang, 2003). The NNAR(p,P,k)_m model uses lagged values as inputs to a feed-forward neural network (Zhang, 2003; Hyndman & Athanasopoulos, 2021):

$$y_t = f(y_{t-1}, y_{t-2}, \dots, y_{t-p}, t_{t-m}, \dots, y_{t-pm}) + \varepsilon_t \quad (7)$$

where p is the number of non-seasonal lags; P is the number of seasonal lags; m is the seasonal period ($m = 12$ for monthly data); $f(\cdot)$ is a nonlinear logistic sigmoid activation function applied through a single hidden layer; ε_t is the error term; and t denotes the time period. All inputs are scaled to $[0,1]$ prior to training (Zhang, 2003; Hyndman & Athanasopoulos, 2021).

These models have been successfully applied in forecasting CPI, agricultural commodity prices, and inflation in various countries (Chi & Chi, 2021; Giri & Giri, 2024; Kmytiuk et al., 2024; Özden, 2023; Vibas & Raqueño, 2019; Xia, 2024). However, to the best of the authors' knowledge, their comparative performance for the highly seasonal and volatile fruits and vegetables category in Albania remains underexplored.

The main objectives of this study are to: identify the optimal forecasting model among SARIMA, ETS, and NNAR for the Albanian CPI of fruits and vegetables; evaluate their accuracy using MAPE and ACF1 of residuals; and analyze forecast uncertainty through 95% confidence intervals.

While prior studies examined forecasting models for the overall Albanian CPI (Gjika Dharmo et al., 2018), no study

specifically focused on the fruits and vegetables subcategory, representing a clear research gap that this study sought to address. This gap was particularly notable given the economic significance of this category in the Albanian household consumption basket. Against this background, the following hypothesis is proposed:

H1: *The SARIMA model will achieve superior forecasting accuracy compared to ETS and NNAR for the Albanian CPI of fruits and vegetables, due to the dominant seasonal structure of the series.*

This hypothesis is grounded in the established strength of SARIMA models in capturing regular seasonal cycles in agricultural price data (Chi & Chi, 2021; Kmytiuk et al., 2024). While NNAR is capable of detecting nonlinear patterns, its advantage is expected to be limited given the dominance of seasonality over nonlinear dynamics in this context (Zhang, 2003). ETS is expected to perform adequately but with higher error margins compared to SARIMA (Hyndman et al., 2008). The validity of this hypothesis is tested through a rigorous comparative evaluation of all three models using out-of-sample forecast accuracy metrics (Makridakis et al., 1998).

Literature Review

Forecasting agricultural product prices and food CPI sub-indices has attracted significant attention due to strong seasonality and volatility. Recent studies mainly compare traditional statistical models (SARIMA, ETS, ARIMA) with machine learning techniques (NNAR, LSTM, Prophet) using monthly data. Kmytiuk et al. (2024) found that SARIMA performed best for capturing seasonality in Ukrainian vegetable prices, while ETS excelled in short-term forecasts. Xia (2024) and Giri and Giri (2024) reported that ARIMA/SARIMA outperformed LSTM and Prophet for vegetable prices in Nepal, showing better out-of-sample accuracy and interpretability. Liu (2024) demonstrated that a hybrid ARIMA-LSTM model improved CPI forecasting accuracy in China by combining linear and nonlinear components.

Özden (2023) showed that ARIMA achieved the lowest RMSE for the Turkish Agricultural Input Price Index compared to CNN and LSTM, suggesting that traditional statistical models may outperform deep learning approaches in short monthly agricultural series. This finding is relevant to the present study, particularly regarding the limited data span characterizing the Albanian CPI for fruits and vegetables. Similar results were obtained by Udari and Hemachandra (2024) in Sri Lanka and Chi and Chi (2021) for global banana prices, where statistical models often proved more robust than neural networks for seasonal series with limited observations. Advanced hybrids such as STL-ATTLLSTM (Yin et al., 2020) and ARIMAX (Vibas & Raqueño, 2019) further improved performance by addressing seasonality and commodity interdependencies.

Overall, the literature indicates that SARIMA and ETS models are reliable and interpretable for seasonal agricultural prices, while machine learning models can capture nonlinearity but frequently require larger datasets. However, most studies focus on individual commodities rather than aggregated CPI sub-groups, and very few examine the Albanian context with its high supply-side volatility in fruits and vegetables.

Method

Dataset

The data were sourced from the Albanian Institute of Statistics (INSTAT) in CSV format, covering monthly CPI observations from January 2007 to December 2025, structured according to the COICOP classification. The dataset comprised 12 rows and 229 columns, representing food groups and monthly periods respectively, with all values expressed as price indices relative to the INSTAT- defined base year. Standard preprocessing was applied, including removal of redundant column prefixes, (retaining only year-month identifiers (e.g., 2007-01) replacement of Albanian-specific characters (ë, ç) for software compatibility, and reshaping from wide to long format using `pivot_longer()` in R, yielding 2,508 observations across three columns: food category, month, and price index value.

This study focuses on two time series, fruits and vegetables - extracted from the "Food" category within "Food and Non-Alcoholic Beverages". The vegetables series, originally labeled "*zarzavate_perfshire_patate*" (vegetables including potatoes) in the source data, was renamed for clarity, with no effect on the underlying data. All data processing and analysis were conducted using the statistical programming language R (R Core Team, 2024). The following packages were employed: *tidyr* for data reshaping and transformation from wide to long format (Wickham et al., 2023); *tseries* for stationarity testing, specifically the ADF and KPSS procedures; *forecast* for time series modelling and forecasting, including the implementation of SARIMA, ETS, and NNAR models (Hyndman & Khandakar, 2008) and *ggplot2* for data visualization and graphical representation of the time series, decomposition components, and forecast outputs (Wickham, 2016).

Analytical Framework

For each series, exploratory analysis began with visual inspection of the time series plot to identify the presence of trend, seasonality, and cyclical components. Classical additive decomposition was subsequently applied to isolate the trend, seasonal, and irregular components, providing a structured basis for subsequent statistical testing.

Stationarity was assessed using two complementary tests: the ADF test (Dickey & Fuller, 1979), which evaluates the null hypothesis of a unit root (non-stationarity), and the KPSS test (Kwiatkowski et al., 1992), which evaluates the null hypothesis of stationarity. Both tests were applied at a significance level of $\alpha = 0.05$ (95% confidence level). A series was considered stationary when the ADF test rejected the null hypothesis of a unit root (p-value < 0.05) and the KPSS test simultaneously failed to reject the null hypothesis of stationarity (p-value > 0.05). Using both tests conjointly reduce the risk of contradictory conclusions that may arise when relying on a single procedure (Kwiatkowski et al., 1992).

Where the original series was found to be non-stationary, first differencing was applied and both tests were repeated on the differenced series. ACF and PACF plots of the differenced series were examined to guide model identification.

For pure AR processes, the PACF cuts off sharply after lag p while the ACF decays gradually; for pure MA processes, the pattern is reversed (Box et al., 2015). In practice, however, mixed ARMA processes produce gradual decay in both functions, making precise order identification unreliable through visual inspection alone (Box et al., 2015; Enders, 2014). Accordingly, ACF and PACF plots were used as exploratory tools only, and final model specification was determined by the `auto.arima()` function, which systematically selects the optimal SARIMA structure by minimizing AICc (Hyndman & Khandakar, 2008).

Three forecasting models were applied to each series: SARIMA selected via Auto ARIMA, incorporating autoregressive, moving average, and seasonal components, with a drift term included where a long-term trend was identified. ETS fitted using the `ets()` function in R, which automatically selects the optimal combination of additive or multiplicative components by minimizing AICc. NNAR an ensemble-based model averaged over 20 networks, capable of capturing non-linear and seasonal patterns in the data.

Residual diagnostics were performed for each model, including visual inspection of residual plots and the Ljung-Box test to verify white noise behavior. Model performance was evaluated and compared using the following accuracy metrics: RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), MAPE (Mean Absolute Percentage Error), MASE (Mean Absolute Scaled Error), and ACF1 (first-order residual autocorrelation). The model achieving the lowest error metrics was selected as the preferred forecasting model for each series.

The selected models were used to generate 12-month ahead point forecasts for 2026, accompanied by 95% prediction intervals. All three models were additionally compared in terms of forecast values and interval width to provide a comprehensive assessment of forecast uncertainty. It should be noted that all forecasts assume structural stability of the series and do not account for unexpected macroeconomic or climatic shocks.

Result

The monthly CPI series for fruits and vegetables over the period 2007- 2025 are presented in Figures 1(a) and (b), respectively. The Fruit CPI exhibited a clear long-term upward trend, strong 12-month seasonality, and a modest cyclical component, with a notable acceleration in price growth observed after 2015. Similarly, the Vegetable CPI revealed a sustained upward trend, with price growth accelerating significantly after 2014. Both series exhibited robust annual seasonality characterized by recurring intra-annual fluctuations; however, the amplitude of seasonal variation intensified more markedly in the vegetable series during the latter years of the study period.

These structural features confirmed that both series were non-stationary in level and necessitated the application of advanced time series modelling techniques to ensure accurate forecasting.

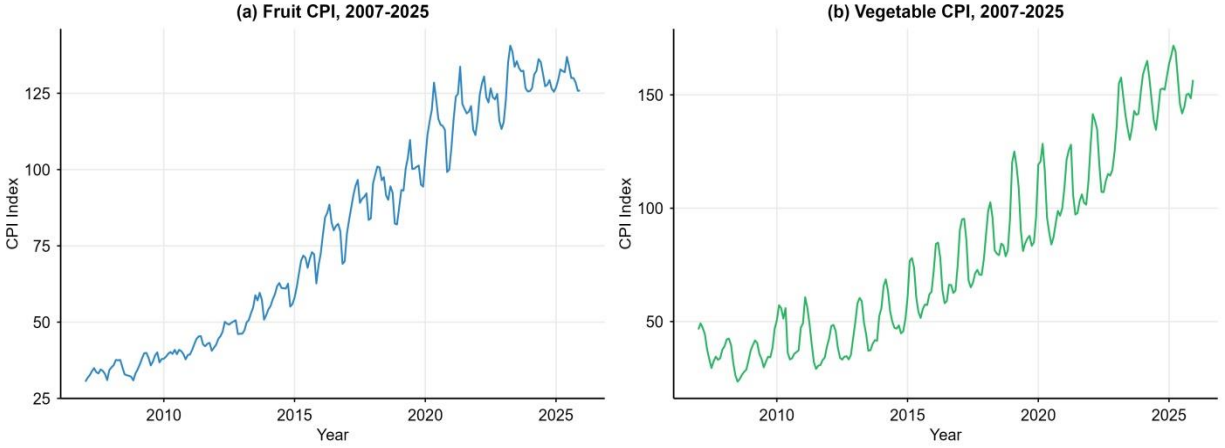


Figure 1. Monthly CPI for fruits and vegetables, 2007-2025

Classical additive decomposition was applied to both series, expressing each as $Y_t = T_t + S_t + R_t$ (Makridakis et al., 1998). The additive specification was selected given that the amplitude of seasonal fluctuations remained approximately constant over time in both cases (Figure 2).

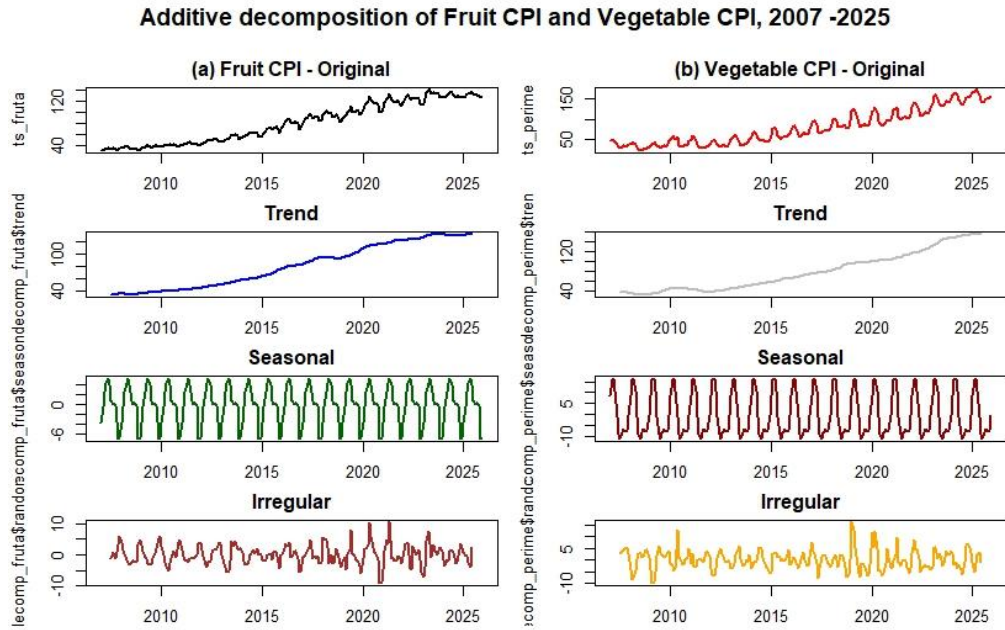


Figure 2. Additive decomposition of Fruit CPI and Vegetable CPI

For the Fruit CPI, the trend component confirmed a gradual long-run price increase, accelerating after 2015. The seasonal component displayed a stable 12-month pattern, while the irregular component fluctuated randomly around zero with no discernible structure. For the Vegetable CPI, the trend confirmed a continuous price increase, with a steeper rise after 2020. The seasonal component similarly reflected a regular annual cycle, with prices declining during

harvest months and rising in winter. The irregular component revealed several outlier observations after 2021, attributable to unexpected market shocks. Overall, both series confirmed the presence of a significant trend and well-defined seasonality, indicating non-stationarity in level and the need for further transformation prior to modelling.

The stationarity of both series was assessed using the ADF and KPSS tests, which complement each other: the ADF tests the null hypothesis of a unit root, while the KPSS tests the null hypothesis of stationarity. Results for both series are summarized in Table 1. For the Fruit CPI, the ADF test at level yielded a statistic of -2.845 ($p = 0.221$), failing to reject the null hypothesis of non-stationarity, while the KPSS statistic of 4.613 ($p = 0.01$) rejected the null of stationarity. Following first differencing, the ADF statistic fell to -9.269 ($p = 0.01$) and the KPSS to 0.033 ($p = 0.1$), confirming stationarity. The series was classified as $I(1)$.

For the Vegetable CPI, the ADF test at level yielded -3.900 ($p = 0.015$), nominally suggesting stationarity. However, the KPSS statistic of 4.274 ($p = 0.01$) rejected the null of stationarity, indicating the presence of a deterministic trend. Following first differencing, the KPSS statistic declined to 0.045 ($p = 0.1$), confirming stationarity. The series was accordingly also classified as $I(1)$.

Table1. ADF and KPSS results, Fruit and Vegetable CPI

Test	Series	At Level	First Difference	Order
ADF	Fruit CPI	-2.845 ($p=0.221$)	-9.269 ($p= 0.01$)	$I(1)$
KPSS	Fruit CPI	4.613 ($p=0.01$)	0.033 ($p = 0.1$)	$I(1)$
ADF	Vegetable CPI	-3.900 ($p = 0.015$)	-	-
KPSS	Vegetable CPI	4.274 ($p = 0.01$)	0.045 ($p= 0.1$)	$I(1)$

Examination of the ACF and PACF plots of both differenced series revealed significant spikes at seasonal lag 12, confirming strong 12-month seasonality, alongside mixed ARMA dynamics in both functions. Given the complexity of visual identification in such cases, final model specification was determined via `auto.arima()`, as outlined in the Methodology(Method). The Auto ARIMA procedure was applied to both series to identify the optimal SARIMA specification. For the Fruit CPI, the selected model was $SARIMA(1,0,0)(0,1,2)[12]$ with drift, while for the Vegetable CPI the procedure identified $SARIMA(2,1,1)(0,1,1)[12]$ without drift.

For the Fruit CPI, the estimated coefficients were $AR(1) = 0.896$, $SMA(1) = -0.473$, $SMA(2) = -0.244$, and drift (δ) = 0.496 , with $\sigma = 2.762$, $AIC = 1066.52$, $AICc = 1066.81$ and $BIC = 1083.40$. For the Vegetable CPI, the estimated coefficients were $AR(1) = 0.696$, $AR(2) = -0.140$, and $SMA(1) = -0.639$, with $\sigma = 3.842$, $AIC = 1201.48$, $AICc = 1201.77$ and $BIC = 1218.33$. The inclusion of a drift term in the Fruit CPI model reflected the presence of a statistically significant long-run upward trend, whereas the Vegetable CPI series did not exhibit a sufficiently persistent linear trend to warrant its inclusion. The estimated coefficients for both models are reported in Table 2.

Table 2. SARIMA estimation and forecast accuracy results

Model	RMSE	MAE	MAPE(%)	ACF1	AIC	BIC	sigma	drift
SARIMA(1,0,0)(0,1,2)[12] with drift (Fruit)	2.663	1.895	2.38%	0.021	1066.52	1083.4	2.762	0.496
SARIMA(2,1,1)(0,1,1)[12] (Vegetable)	3.696	2.632	3.76%	-0.018	1201.48	1218.33	3.842	-

Residual diagnostics for both SARIMA models are presented in Figures 3 and 4 respectively.

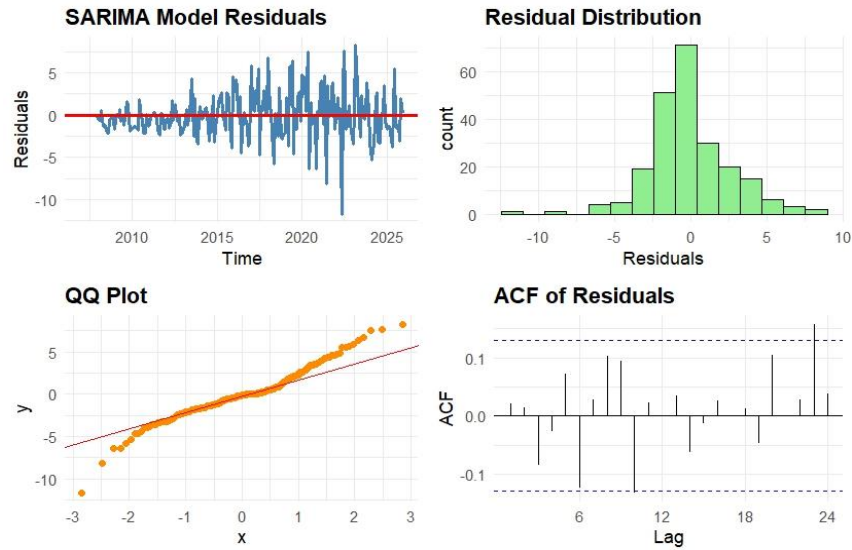


Figure 3. Residual diagnostics for SARIMA(1,0,0)(0,1,2)[12] - Fruit CPI

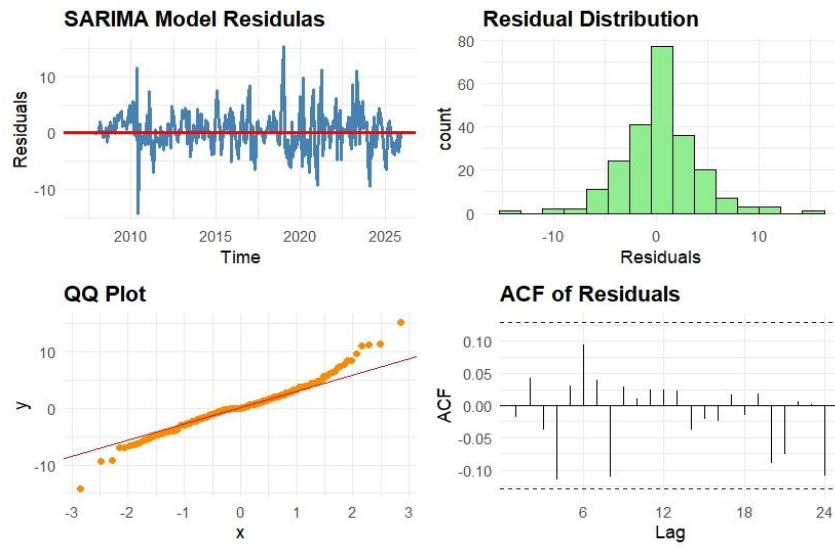


Figure 4. Residual diagnostics for SARIMA(2,1,1)(0,1,1)[12] - Vegetable CPI

For both series, residuals fluctuated randomly around zero with no systematic trend or seasonal structure, and histograms indicated approximate normality, with minor deviations in the tails attributable to isolated outlier observations. The ACF plots showed no statistically significant autocorrelation at any lag. The Ljung-Box test formally confirmed white noise behaviour for both models: Fruit CPI, $Q^* = 27.775$ ($df = 21$, $p = 0.147$); Vegetable CPI, $Q^* = 17.732$ ($df = 20$, $p = 0.605$), failing to reject the null hypothesis of no residual autocorrelation at the 5% significance level.

Following residual validation, 12-month ahead forecasts for 2026 were generated using the estimated SARIMA models. The forecast plots are presented in Figure 5, while the numerical values are reported in Table 3. For the Fruit CPI, the index was projected to rise from 129.48 in January to a seasonal peak of 143.04 in June, before moderating toward year-end (December: 133.46). The 95% prediction intervals widened progressively over the horizon, from [124.07, 134.89] in January to [121.73, 145.19] in December.

For the Vegetable CPI, the forecast indicated a seasonal peak in March at 178.38 [166.88, 189.87], followed by a summer decline to a minimum of 148.43 in July [134.52, 162.35], consistent with the harvest season effect identified in the decomposition analysis. Price levels subsequently recovered toward December (166.18 [150.05, 182.30]). The wider prediction intervals relative to the Fruit CPI reflected the greater inherent volatility of vegetable prices. In both series, seasonal patterns were preserved throughout the forecast horizon.

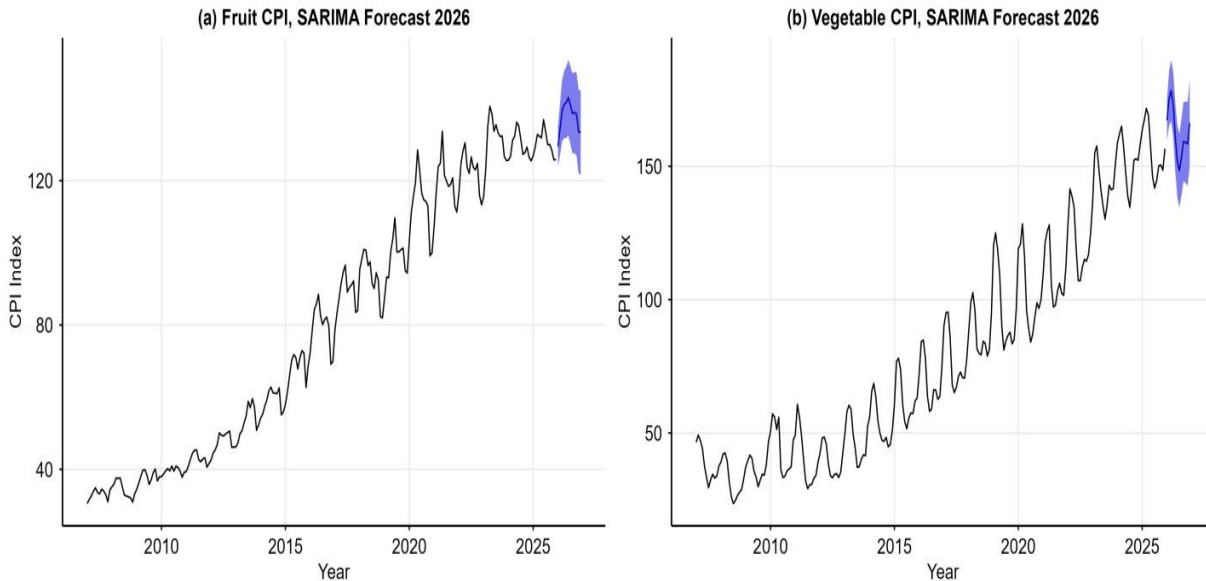


Figure 5. SARIMA forecasts, Fruit and Vegetable CPI 2026

Table 3. SARIMA forecasts and 95% prediction intervals, 2026

Month	Fruit CPI			Vegetable CPI		
	Point Forecast	Lo 95%	Hi 95%	Point Forecast	Lo 95%	Hi 95%
January	129.48	124.07	134.89	167.18	159.65	174.71
February	134.30	127.04	141.57	175.28	165.09	185.46
March	139.39	130.93	147.86	178.38	166.88	189.87
April	141.08	131.76	150.39	173.31	161.01	185.61
May	141.70	131.75	151.64	161.23	148.33	174.13
June	143.04	132.62	153.46	151.79	138.37	165.22
July	140.95	130.15	151.74	148.43	134.52	162.35
August	138.62	127.54	149.70	153.25	138.87	167.64
September	138.90	127.59	150.20	159.28	144.44	174.12
October	138.42	126.94	149.90	158.97	143.69	174.26
November	133.50	121.88	145.12	158.43	142.72	174.14
December	133.46	121.73	145.19	166.18	150.05	182.30

The ETS model was estimated using the `ets()` function in R, which automatically selected the optimal specification by minimizing the AICc criterion. The procedure identified $ETS(M,Ad,M)$ for both the Fruit and Vegetable CPI series, indicating multiplicative errors, a damped additive trend, and multiplicative seasonality. The selection of multiplicative components for both error and seasonality terms was consistent with the increasing variance observed across the sample period. The estimated parameters and fit statistics for both models are reported in Table 4 and 5.

Table 4. ETS smoothing parameters, Fruit and Vegetable CPI

Model	$\alpha(\text{level})$	$\beta(\text{trend})$	$\gamma(\text{seasonal})$	$\phi(\text{damping})$	$\sigma(\text{sigma})$
ETS(Fruit)	0.835	0.014	0.001	0.98	0.033
ETS(Vegetable)	0.999	0.01	0.001	0.911	0.056

Table 5. ETS forecast accuracy, Fruit and Vegetable CPI

Model	AIC	AICc	BIC	RMSE	MAE	MAPE(%)	ACF1
ETS(Fruit)	1643.543	1646.816	1705.272	2.758	1.981	2.44%	0.074
ETS(Vegetable)	1873.639	1876.912	1935.367	4.37	3.062	3.95%	0.171

For the Fruit CPI, the estimated parameters indicated rapid level adjustment ($\alpha = 0.835$), a slow and stable trend evolution ($\beta = 0.014$), a highly stable seasonal pattern ($\gamma = 0.001$), and gradual trend attenuation ($\phi = 0.98$). For the

Vegetable CPI, the near-unity level parameter ($\alpha = 0.999$) reflected the high volatility of the series, while the low trend and seasonality parameters ($\beta = 0.01$, $\gamma = 0.001$) confirmed stable long-run dynamics. The damping parameter ($\phi = 0.911$) prevented unstable long-run projections.

Both models demonstrated satisfactory in-sample fit, with MAPE values of 2.44% and 3.95% for the Fruit and Vegetable CPI respectively. The validated ETS models were used to generate 12-month ahead forecasts for 2026, presented in Figure 6 and Table 6. For the Fruit CPI, the index was projected to rise from 131.37 in January to a seasonal peak of 146.91 in May, before declining toward year-end (November: 126.56). The prediction intervals widened progressively from [122.91, 139.84] in January to [101.64, 153.94] in December. For the Vegetable CPI, the forecast indicated peak values in February and March (192.20 and 192.32), followed by a pronounced summer decline to 131.31 in July [91.70, 170.91], before recovering toward December (156.49 [93.48, 219.49]).

The widening prediction intervals across both series reflected the inherent uncertainty associated with longer forecast horizons, a pattern commonly observed in multiplicative ETS models applied to volatile agricultural price data. This uncertainty was particularly pronounced for the Vegetable CPI, where the broader intervals toward year-end were consistent with the higher volatility and less predictable seasonal dynamics characterizing this series. Nevertheless, the central forecast trajectories remained economically interpretable and aligned with the established seasonal patterns observed in the historical data.

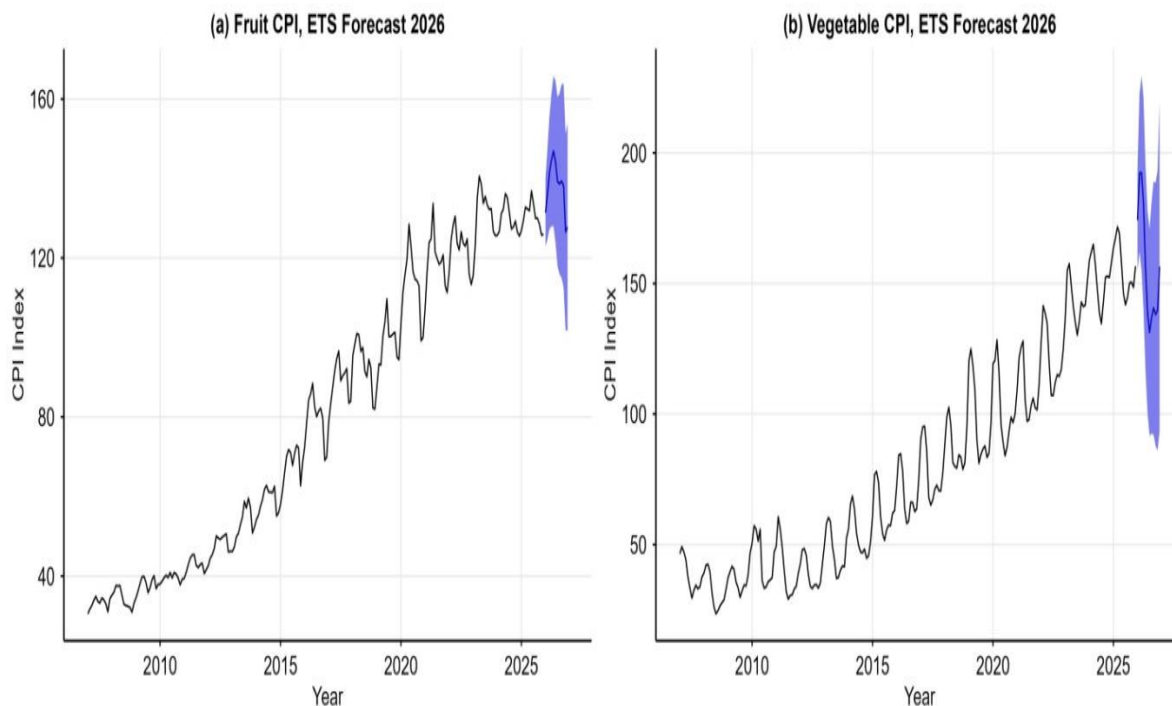


Figure 6. ETS forecasts, Fruit and Vegetable CPI, 2026

Table6. ETS forecasts and 95% prediction intervals, 2026.

Month	Fruit CPI			Vegetable CPI		
	Point Forecast	Lo 95%	Hi 95%	Point Forecast	Lo 95%	Hi 95%
January	131.37	122.91	139.83	174.27	155.0	193.54
February	136.22	124.71	147.72	192.20	161.97	222.42
March	141.62	127.32	155.93	192.32	155.07	229.56
April	144.71	127.98	161.44	180.41	139.85	220.96
May	146.91	127.97	165.85	155.69	116.37	195.02
June	144.16	123.79	164.53	138.97	100.34	177.6
July	139.18	117.89	160.47	131.31	91.70	170.91
August	138.66	115.91	161.41	136.73	92.46	181.0
September	139.33	114.99	163.68	140.60	92.11	189.08
October	138.24	112.66	163.81	138.13	87.73	188.54
November	126.56	101.89	151.24	139.72	86.05	193.39
December	127.79	101.64	153.94	156.49	93.48	219.49

The NNAR model was estimated using the `nnetar()` function in R, which selected NNAR(1,1,2)[12] for both series. The model comprised an ensemble of 20 neural networks with a [2–2–1] architecture, incorporating one non-seasonal lag, one seasonal lag ($s = 12$), and two nodes in the hidden layer. Accuracy metrics for both series are reported in Table 7.

Table 7. NNAR model accuracy metrics, Fruit and Vegetable CPI

Model	RMSE	MAE	MAPE(%)	ACF1	$\sigma(\text{sigma})$
NNAR(Fruit)	3.237	2.305	2.89%	0.4	3.24
NNAR(Vegetable)	5.127	3.889	5.63%	0.579	5.13

For the Fruit CPI, the model achieved RMSE = 3.237, MAE = 2.305, and MAPE = 2.89%, indicating satisfactory forecast accuracy. For the Vegetable CPI, the model yielded higher error values (RMSE = 5.127, MAPE = 5.63%), suggesting greater difficulty in capturing the more volatile dynamics of vegetable prices.

The NNAR models were used to generate 12-month ahead forecasts for 2026, with prediction intervals produced via bootstrap simulation. The forecast plots are presented in Figure 7, while the numerical values are reported in Table 8. For the Fruit CPI, the index was projected to rise gradually from 128.04 in January to a seasonal peak of 133.43 in June, before stabilising toward year-end (December: 129.79). The prediction intervals ranged from [121.57, 134.90] in January to [122.89, 136.67] in December.

For the Vegetable CPI, the forecast indicated a gradual increase in the first months of the year, peaking in April at 168.70 [157.42, 179.47], followed by a summer decline to a minimum of 154.97 in August [144.22, 167.06], before recovering toward December (160.73 [150.23, 171.44]).

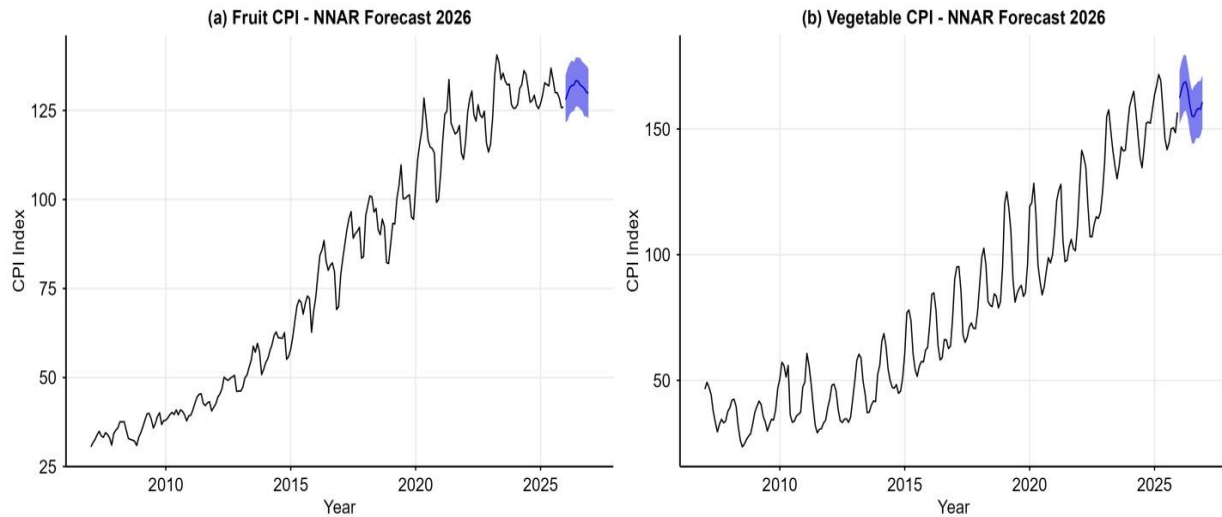


Figure 7. NNAR 12-month forecasts, Fruit and Vegetable CPI, 2026

Table 8. NNAR forecasts and 95% prediction intervals, 2026

Month	Fruit CPI			Vegetable CPI		
	Point Forecast	Lo 95%	Hi 95%	Point Forecast	Lo 95%	Hi 95%
January	128.04	121.57	134.90	162.41	152.16	173.28
February	129.74	122.23	136.89	165.85	154.19	176.97
March	131.40	123.88	138.28	168.36	156.61	179.43
April	132.00	124.68	139.11	168.70	157.42	179.47
May	132.16	124.81	138.60	165.19	154.13	174.96
June	133.43	126.15	139.91	159.20	148.22	169.07
July	133.20	126.10	139.84	155.16	144.21	165.29
August	132.19	125.38	139.79	154.97	144.22	167.06
September	131.76	124.89	138.70	157.20	146.47	167.94
October	131.18	123.36	138.19	158.23	146.22	169.08
November	130.21	123.33	137.64	157.81	147.16	169.05
December	129.79	122.89	136.67	160.73	150.23	171.44

Discussion

Model Comparison

The comparative analysis of the three forecasting models revealed consistent performance differences across both series (Table 9). For the Fruit CPI, SARIMA achieved the lowest MAPE (2.38%) and the ACF1 value closest to zero (0.021), indicating both superior forecast accuracy and the absence of residual autocorrelation. ETS performed comparably (MAPE = 2.44%, ACF1 = 0.074), while NNAR yielded the weakest results (MAPE = 2.89%, ACF1 = 0.4), suggesting that residual autocorrelation remained present in the neural network model.

A similar ranking was observed for the Vegetable CPI, where SARIMA again achieved the best accuracy (MAPE = 3.76%, ACF1 = -0.018), followed by ETS (MAPE = 3.95%, ACF1 = 0.171) and NNAR (MAPE = 5.63%, ACF1 = 0.579). The consistently higher ACF1 values of the NNAR model across both series suggested that it failed to fully capture the temporal dependence structure of the data.

These findings align with existing literature suggesting that SARIMA models tend to outperform machine learning approaches when applied to relatively short and strongly seasonal time series. The limited volume of monthly observations may have constrained NNAR's capacity to generalize effectively, a known challenge in neural network-based forecasting (Hyndman, R.J., & Athanasopoulos, G. (2021).

Table 9. Forecast accuracy comparison, all models and series

Modeli	MAPE(%)	ACF1 - Fruitt	MAPE(%)	ACF1
	Fruit		Vegetable	Vegetable
SARIMA	2.38%	0.021	3.76%	-0.018
ETS	2.44%	0.074	3.95%	0.171
NNAR	2.89%	0.4	5.63%	0.579

Across all three models, the Vegetable CPI consistently yielded higher forecast errors than the Fruit CPI, reflecting the greater inherent volatility of vegetable prices. This finding is consistent with the structural characteristics identified in the decomposition analysis, where the amplitude of seasonal fluctuations was found to increase over time for vegetables but remained stable for fruits. The wider prediction intervals generated for the Vegetable CPI further confirmed that vegetable prices are more sensitive to supply-side shocks, including weather conditions, harvest cycles, and input cost pressures.

Consistency with Existing Literature

The superiority of SARIMA in forecasting agricultural price indices is consistent with findings reported in the broader time series forecasting literature. Several studies have demonstrated that SARIMA models perform competitively in monthly CPI forecasting contexts, particularly when the series exhibits stable seasonal patterns and a well-defined

trend structure (Chi & Chi, 2021; Kmytiuk et al., 2024; Vibas & Raqueño, 2019). The relatively modest performance of the NNAR model in this study may be attributable to the limited sample size, as neural network models typically require larger datasets to fully exploit their capacity for capturing nonlinear dynamics (Zhang, 2003).

Conclusion

This study applied three time series forecasting models - SARIMA, ETS, and NNAR - to monthly CPI data for fruits and vegetables in Albania over the period 2007-2025, with the aim of generating reliable 12-month ahead forecasts for 2026 and evaluating comparative model performance. The results indicated that SARIMA consistently outperformed both ETS and NNAR across all accuracy metrics for both series, achieving the lowest MAPE values of 2.38% and 3.76% for the Fruit and Vegetable CPI respectively. The ETS model performed comparably, while the NNAR model yielded the weakest results, particularly for the Vegetable CPI, where residual autocorrelation suggested incomplete capture of the series dynamics.

The 2026 forecasts confirmed the continuation of upward price trends in both categories, with vegetable prices exhibiting greater seasonal amplitude and forecast uncertainty relative to fruit prices. These findings contribute to the empirical literature on agricultural price forecasting in small open economies, demonstrating that classical linear models retain strong predictive performance even in the presence of evolving seasonal dynamics (Box et al., 2015; Chi & Chi, 2021; Kmytiuk et al., 2024).

Recommendations

Based on the findings, the SARIMA model is recommended as the preferred approach for short-term forecasting of food CPI components in Albania, given its superior accuracy and well-behaved residuals across both series (Box et al., 2015; Chi & Chi, 2021; Kmytiuk et al., 2024).). Statistical institutions should consider integrating such models into regular price monitoring frameworks, as the 12-month ahead forecasts demonstrated reliable predictive capacity for both fruit and vegetable price dynamics (Hyndman & Athanasopoulos, 2021).

From a policy perspective, the upward price trends identified in both series warrant attention from monetary and food policy authorities. Since inflationary pressures in these categories are unlikely to self-correct in the short run, targeted interventions - such as supply chain improvements, agricultural subsidies, or import diversification , may be necessary (Makridakis et al., 1998). Future research should consider incorporating exogenous variables such as energy prices, exchange rates, and import volumes (Hamilton, 1994; Enders, 2014), as well as multivariate models such as VAR or ARIMAX, which could improve forecast accuracy particularly for the more volatile vegetable CPI series (Zhang, 2003; Hyndman & Athanasopoulos, 2021).

References

- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time series analysis: Forecasting and control* (5th ed.). Wiley.
- Chi, Y. N., & Chi, O. (2021). Modeling and forecasting of monthly global price of bananas using seasonal ARIMA and multilayer perceptron neural network. *Econometrics. Ekonometria. Advances in Applied Data Analysis*, 25(3), 22–41. <https://doi.org/10.15611/eada.2021.3.02>
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.
- Enders, W. (2014). *Applied econometric time series* (4th ed.). Wiley.
- Giri, A., & Giri, V. R. (2024). Forecasting cauliflower prices in Nepal: A comparative analysis using seasonal time series and nonlinear models. *ResearchGate*, 1-16. <https://www.researchgate.net/publication/379829420>
- Gjiko Dharmo et al., (2018). Forecasting consumer price index(CPI) using time series models and multi regression models(Albania case study) <https://etalpykla.vilniustech.lt/handle/123456789/154207>
- Hamilton, J. D. (1994). *Time series analysis*. Princeton University Press.
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, 27(3), 1-22. <https://doi.org/10.18637/jss.v027.i03>
- Hyndman, R. J., Koehler, A. B., Ord, J. K., & Snyder, R. D. (2008). *Forecasting with exponential smoothing: The state space approach*. Springer.
- INSTAT. (2025). *Consumer price index database*. Albanian Institute of Statistics. https://databaza.instat.gov.al:8083/pxweb/sq/DST/START__PR__PRCPI/CPI003/table/tableViewLayout1/
- International Labour Organization. (2004). *Consumer price index manual: Theory and practice*. https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40dgreports/%40stat/documents/presentation/wcms_331153.pdf
- Kmytiuk, T., Majore, G., & Bilyk, T. (2024). Time series forecasting of price of the agricultural products using data science. *Agricultural and Resource Economics: International Scientific E-Journal*, 10(3), 5-33. <https://are-journal.com/are/article/view/906>
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1-3), 159-178. [https://doi.org/10.1016/0304-4076\(92\)90104-Y](https://doi.org/10.1016/0304-4076(92)90104-Y)
- Liu, Y. (2024). Forecasting China's Consumer Price Index (CPI) based on combined ARIMA-LSTM models. *Frontiers in Business, Economics and Management*, 16(2), 264-270. <https://drpress.org/ojs/index.php/fbem/article/view/25626>
- Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297-303.

- Makridakis, S., Wheelwright, S. C., & Hyndman, R. J. (1998). *Forecasting: Methods and applications* (3rd ed.). Wiley.
- Özden, C. (2023). Forecasting agricultural input price index using statistical and deep learning methods. *Turkish Journal of Agriculture - Food Science and Technology*, 11(9), 1751-1755. <https://doi.org/10.24925/turjaf.v11i9.1751-1755.6359>
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Udari, U. D. R., & Hemachandra, S. D. S. (2024). Price behaviour, commercialization opportunities and forecasting future prices of selected vegetables in Sri Lanka. *Journal of Technology and Value Addition*, 6(1), 31-50. <https://www.researchgate.net/publication/386232474>
- Vibas, V. M., & Raqueño, A. R. (2019). A mathematical model for estimating retail price movements of basic fruit and vegetable commodities using time series analysis. *International Journal of Advance Study and Research Work*, 2(7), 1-5. <https://doi.org/10.5281/zenodo.3333529>
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer. <https://ggplot2.tidyverse.org>
- Wickham, H., Vaughan, D., & Girlich, M. (2023). *tidyr: Tidy messy data* (R package version 1.3.0). <https://CRAN.R-project.org/package=tidyr>
- Xia, C. (2024). Comparative analysis of ARIMA and LSTM models for agricultural product price forecasting. *Highlights in Science, Engineering and Technology*, 85, 1032-1040. <https://www.researchgate.net/publication/379531846>
- Yin, H., Jin, D., Gu, Y. H., Park, C. J., Han, S. K., & Yoo, S. J. (2020). STL-ATTTLSTM: Vegetable price forecasting using STL and attention mechanism-based LSTM. *Agriculture*, 10(12), 612. <https://doi.org/10.3390/agriculture10120612>
- Zhang, G. P. (2003). Time series forecasting using a hybrid ARIMA and neural network model. *Neurocomputing*, 50(1-2), 159-175.

Author Information

Ilir Palla

<https://orcid.org/0009-0007-2122-3579>

Fan S. Noli University

11 Rilindasi Boulevard, Korçë

Albania

Contact e-mail: ipalla@unkorce.edu.al

Jasmina Duka

<https://orcid.org/0009-0003-6471-693X>

Fan S. Noli University

11 Rilindasi Boulevard, Korçë

Albania

Contact e-mail: jasminaduka02@gmail.com
